AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-09-01 | Entry window 1 September 2026 (single day) | 33 records in scope, 98 screened out

33 RECORDS IN SCOPE (98 SCREENED OUT)	**9** SUBTOPIC CLUSTERS OF TEN OCCUPIED	**16** EXPOSURE ASSESS. & MODELLING	**1** EFFECT ESTIMATE WITH A RATIO CI	**22** RECORDS WITH NO HEALTH ENDPOINT

Signal of the day

Two independent *Atmosphere* papers, published the same day and citing nothing in common, make the same argument: the averaging interval written into air-quality regulation destroys most of what a high-rate network measures.

Raso et al. ran certified reference instruments alongside calibrated low-cost sensors through two consecutive New Year's Eve firework events. Conventional monitoring detects that an episode happened; it does not characterise it. Minute-resolution PM_{10} reached **215 $\mu g\,m^{-3}$**, roughly three times the corresponding hourly value — a peak amplification factor of **2.90–2.93** that the hourly channel simply does not contain. Mohd Nadzir et al. deploy the complementary case: a single low-cost multi-sensor node in the Antarctic Peninsula, **58,652 valid minute observations** over May–June 2023, in an environment where the daily mean is near-constant and therefore uninformative. At minute resolution the record resolves **142 single-pollutant and 30 multi-pollutant enhancement events**; the median $PM_{2.5}/PM_{10}$ ratio of **0.96** says the mass is essentially all fine, which is the diagnostic that distinguishes camp and vessel combustion from sea-salt and local dust. Both frame the same brief from opposite ends of the concentration range — 215 $\mu g\,m^{-3}$ urban peaks against a 6.3 $\mu g\,m^{-3}$ polar background — and both land on temporal resolution, not accuracy, as the property that justifies the sensor. That is the strongest argument this watch has carried for the distributed low-cost network as an instrument in its own right rather than a cheap approximation to a reference station. Neither reports a drift or co-location series long enough to say what those minute values are worth after six months in the field.

What changed today

- **The largest corpus since 27 August, and the most lopsided.** *Exposure assessment & modelling* takes **16 of 33** records and **22 of 33 (67%) carry no human health endpoint at all**. Seven of the sixteen arrive in one *J. Environ. Sci.* deposit batch, so the cluster's size is partly a publishing artefact, not a shift in the field.
- **A policy evaluation with an individual-level counterfactual.** Sun D et al. (*Lancet Planet Health*) use China Kadoorie Biobank participants (n=34,862) assigned to intervention or control by their municipality's PM-reduction target under the 2013 Clean Air Act. The intervention group's predicted 10-year CVD risk rose **3.95% (95% CI 3.18–4.72) less** than control; per 10 $\mu g\,m^{-3}$, **1.80 (1.34–2.27) percentage points** for $PM_{2.5}$. The outcome is a *predicted* risk score, not incident disease, and assignment is a municipal policy target — but this is the closest thing to a quasi-experiment the PM–CVD literature has at this scale.
- **Only one estimate in the whole corpus is a ratio with a confidence interval.** Gómez-Olarte et al. report **aOR 1.50 (1.02–2.23)** for bronchitis at age one in boys per IQR (2.0 $\mu g\,m^{-3}$) $PM_{2.5}$, with IL-8 up **31.9% (18.6–46.7)** in girls. Today's forest plot is therefore a single point, not a synthesis; the Lancet numbers are percentage-point differences on a risk score and cannot share a log-ratio axis with it.
- **Two records converge on the same conclusion about PM mass as a metric.** Arora et al.'s commentary restates the source-dependence of oxidative potential — biomass-burning particles up to **35×** the OP of suburban $PM_{2.5}$, O_3 aging generally *reducing* it — while Arpornwichanop & Chuersuwan reach the same place from the elemental side: deposited intake in Nonthaburi ranks Al, Fe, B, Zn highest, the solubility- and toxicity-adjusted priority ranks **As, Cd, Sb**.
- **Fine-scale exposure modelling posts a concrete number.** Wang T et al.'s IAQMS-Industry resolves $PM_{2.5}$ at **100 m** inside industrial parks and attributes **22.9–26.4%** of park $PM_{2.5}$ to local point sources against 1.6–13.7% for the same emissions in a 3 km grid, with normalised mean bias improving from −16.9%–−7.7% to **3.1–6.2%**. That gap is the sub-grid smoothing error a dense network would be measuring against.
- **Sub-Saharan Africa and Latin America are both absent again**; China supplies 13 of 33, Europe 6, North America 2. **Source breakages:** OpenAlex HTTP 403 (plan-gated date filter, unchanged since 28 July) and arXiv HTTP 429 on four attempts, so the preprint-instrumentation leg contributed nothing. All three Consensus hits on the sensor-calibration axis were already in the store (2 Aug, 5 Aug, 30 Jul).

Corpus analytics

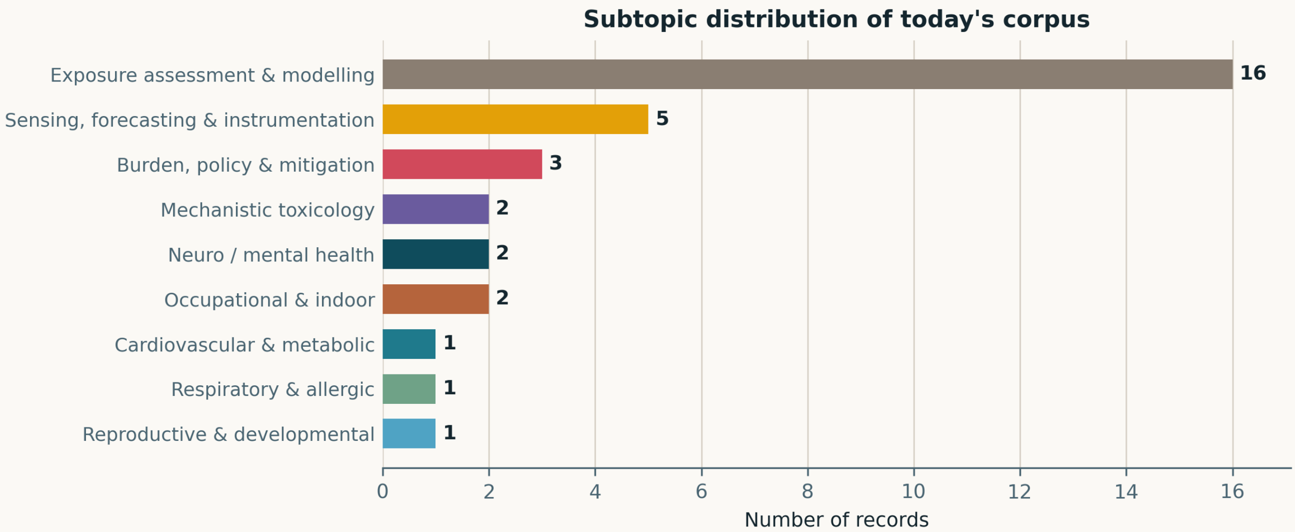
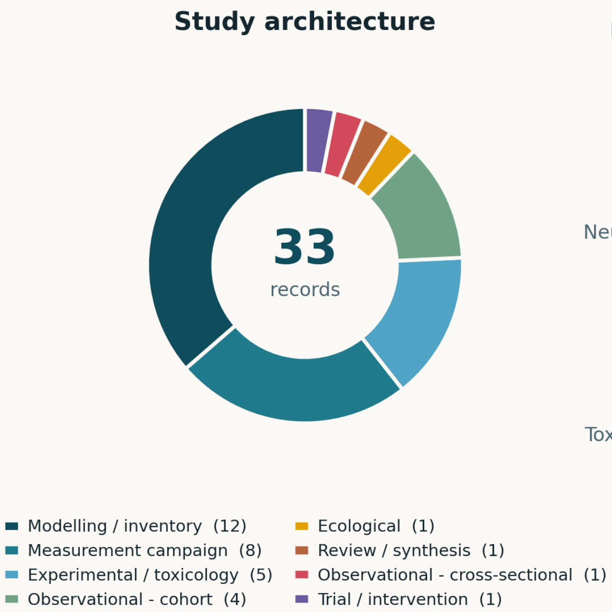


Fig. Subtopic assignment of the 33 in-scope records. Nine of ten clusters are occupied; only *Other clinical endpoints* is empty. *Exposure assessment & modelling* at 16 is the largest single-cluster count in any single-day issue since the 26 July seed issue (26, same cluster), and the seven *J. Environ. Sci.* papers inside it (soot morphology, brown-carbon triplets, aqSOA, NH_3–isoprene, terrain blocking, N-containing organics, industrial-park CTM) are one publisher batch rather than seven independent signals.



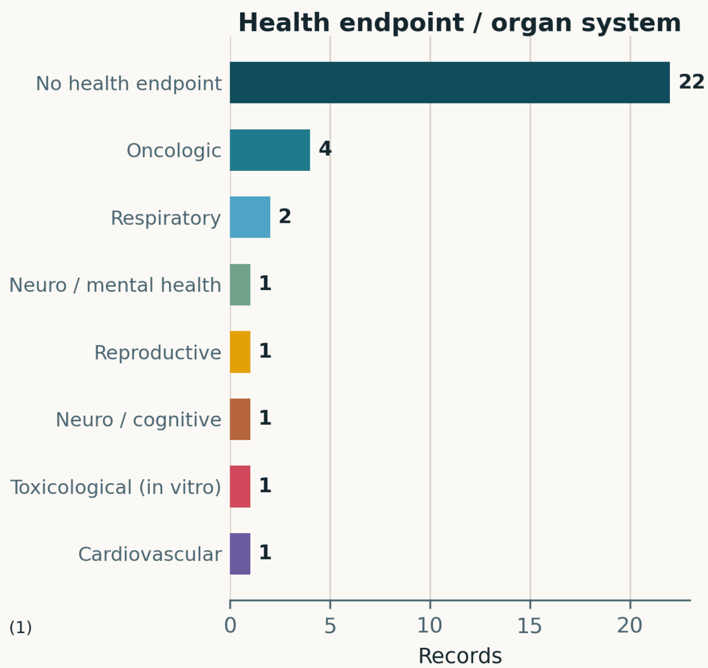


Fig. Left — study architecture. *Modelling / inventory* (12) and *Measurement campaign* (8) together hold 20 of 33; four cohorts, five laboratory/toxicology studies, one cross-sectional, one intervention and one review make up the rest. Right — health endpoint. **Twenty-two of thirty-three records (67%) carry no human health endpoint** — level with 9 August and behind only 20 August (72%), 2 August (71%) and 5 August (70%). Of the eleven that do, four are cancer-risk screens rather than observed incidence.

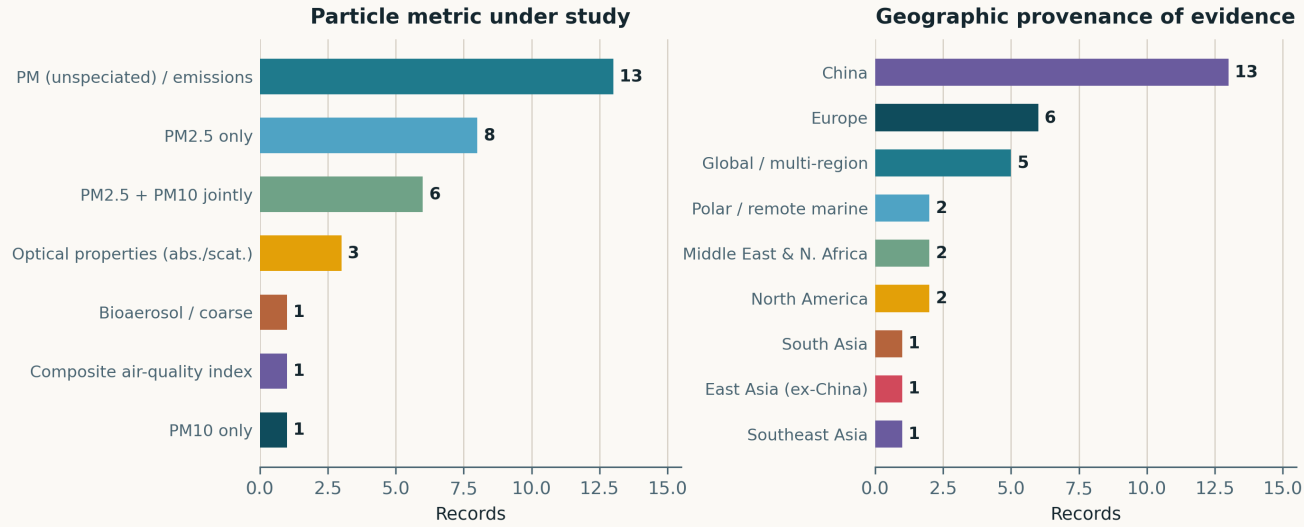


Fig. Left — particle metric. $PM_{2.5}$ alone is the exposure in 8 records and appears jointly with PM_{10} in 6 more; three records are optical (soot morphology, Mediterranean dust, N-containing organic absorbance) and one is bioaerosol. The 13 in *PM (unspeciated) / emissions* are mostly the chemistry and inventory papers. Right — geographic provenance. China at 13 is 39% of the corpus. **Sub-Saharan Africa and Latin America are both absent**, as they were on 31 August; the two *Polar / remote marine* entries are the Antarctic sensor node and the CAFE-Pacific aircraft campaign.

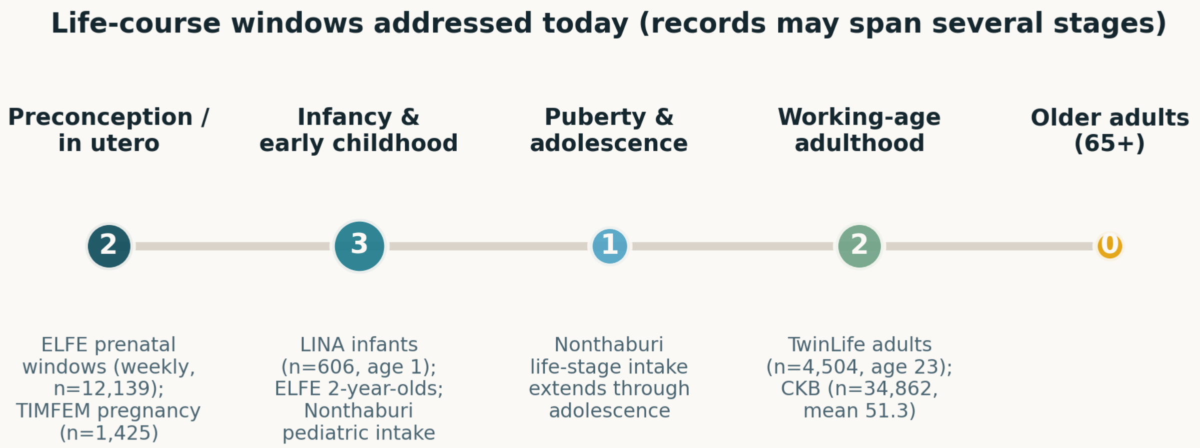


Fig. Life-course windows addressed; counts are non-exclusive and only the six records studying identifiable human populations contribute. The **65+ cell is zero**: no record in this issue reports an age-stratified estimate for older adults, even though the Kadoorie cohort (mean baseline age 51.3, followed to 2020–21) certainly contains a large older stratum. Scoring it otherwise would misdescribe what the paper reports.

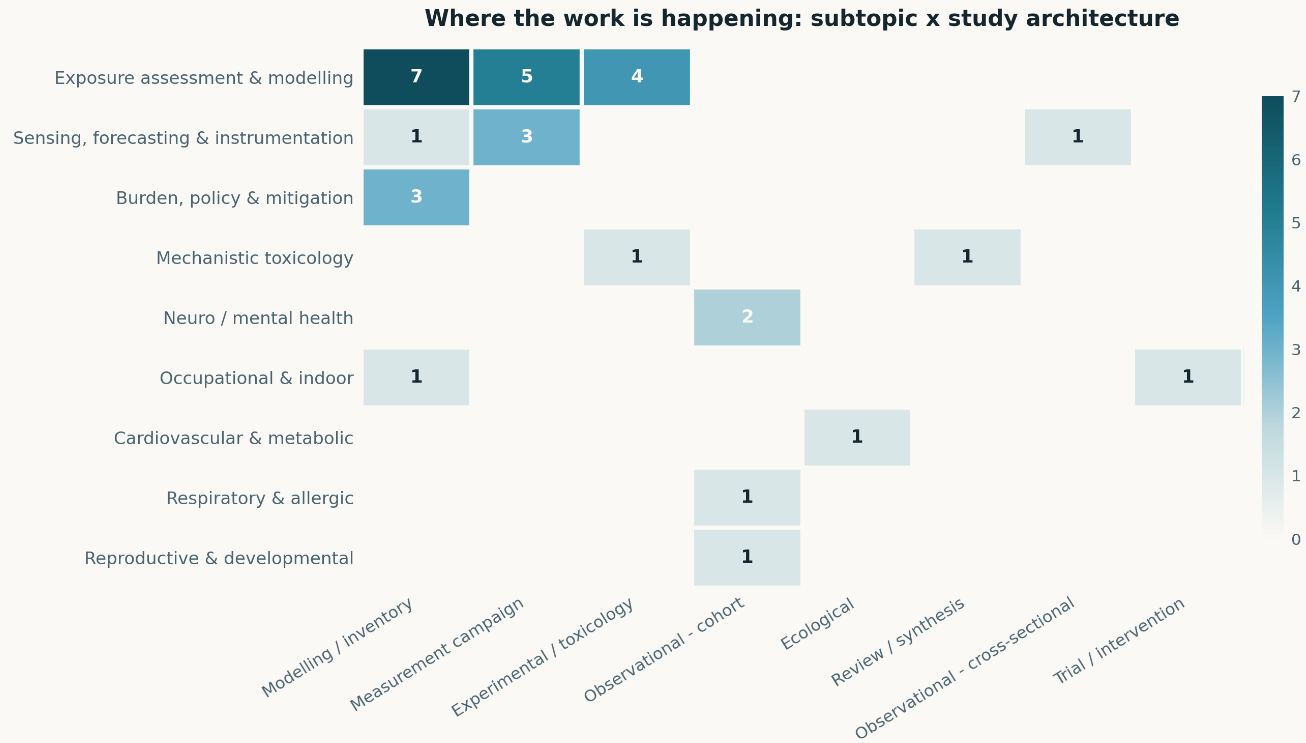


Fig. Subtopic × study-architecture cross-tabulation. The separation this watch keeps flagging is at its widest today: *Exposure assessment & modelling* and *Sensing, forecasting & instrumentation* together hold **21 of 33 records and contribute exactly zero effect estimates**. The four cohort records span only three health clusters — two of them the same one (*Neuro / mental health*) — and the single cardiovascular record is not a cohort but a difference-in-differences design.

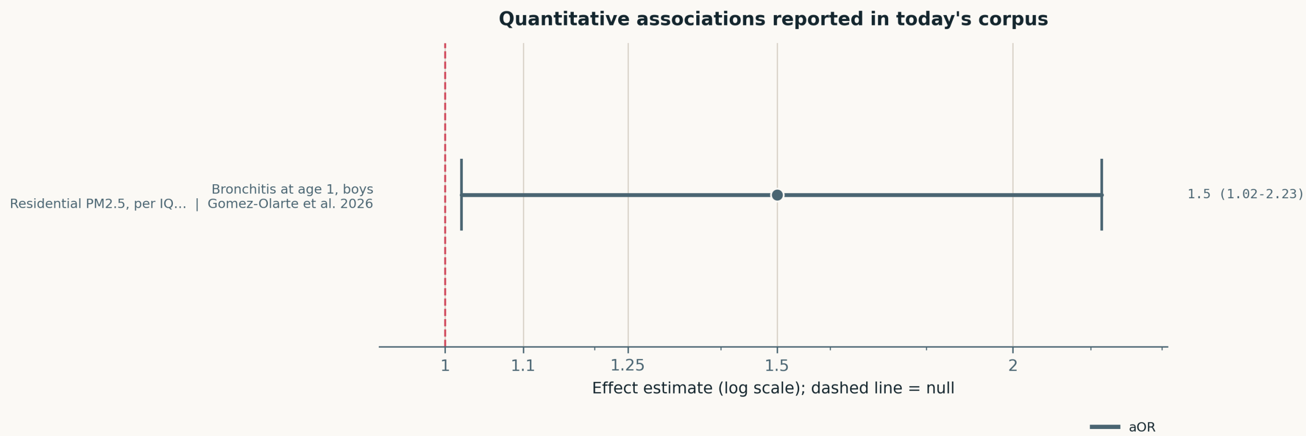


Fig. A single estimate. Only Gómez-Olarte et al. report an effect on a ratio scale with a confidence interval, so this panel displays one contrast and is not a synthesis. The lower bound (1.02) sits against the null, the cohort is small (n=606 for the respiratory outcome), and the estimate is sex-stratified post hoc — read it as consistent with the cytokine gradient in the same paper, not as independent confirmation of it.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Neuro / mental health 2 records

Barbalat et al. 2026 Environ Sci Technol

Heat, cold and ambient pollution in the first 1000 days and autism risk indicators

ELFE national birth cohort, n=12,139 two-year-olds, M-CHAT-R screening score against weekly temperature and $PM_{2.5}/PM_{10}/NO_2$ from 1 km daily predictions (200 m for NO_2 and temperature in large urban areas), modelled with confounder-adjusted distributed lag non-linear models. **Prenatal air pollution was not associated with the screening score**; night-time heat in the early third trimester was, and survived adjustment for $PM_{2.5}$ and PM_{10}. Postnatally the pattern is sex-specific: higher $PM_{2.5}$ and PM_{10} at months 7–11.5 raised scores in males relative to females. The outcome is a screening instrument, not a diagnosis, and the sex-stratified postnatal windows are the kind of finding a DLNM will produce by construction unless pre-registered. Useful as a negative result on the prenatal PM window.

doi: 10.1021/acs.est.5c18062

Kuznetsov et al. 2026 Environ Res

Particulate matter and life satisfaction, across and within twin pairs

TwinLife, n=4,504 young-adult twins (2,249 MZ / 2,255 DZ), mean age 23.0, up to six waves 2014–2022, decomposing the PM–wellbeing association into between- and within-pair components. The effect runs **between families** ($b=-0.05$, SE 0.01, $p<.001$) and strengthens with age and calendar year for PM_{10}. Within pairs, after removing shared family variance, the association is null in DZ twins and only nominally significant in MZ twins for PM_{10} ($b=-0.09$, SE 0.04, p=0.04) and not at all for $PM_{2.5}$ (p=.15). That is the honest reading: most of the apparent PM effect on subjective wellbeing is confounded by family SES and neighbourhood, and the design is what makes that visible. A rare case of a study whose null is the contribution.

doi: 10.1016/j.envres.2026.125565

Cardiovascular & metabolic 1 record

Sun D et al. 2026 Lancet Planet Health

China's Clean Air Act and individual-level cardiovascular risk

China Kadoorie Biobank, n=34,862 CVD-free at baseline, mean age 51.3, split into intervention (n=25,497) and control (n=9,365) by the local government's PM reduction target under the 2013 Clean Air Act, with 10-year predicted CVD risk as the outcome and a difference-in-differences estimator. The intervention group's predicted risk rose **3.95% (3.18–4.72) less** than control, with larger differences under stricter enforcement. Per 10 $\mu g\,m^{-3}$, 2013–2021: $PM_{2.5}$ **1.80 (1.34–2.27)** percentage points, PM_{10} 1.24 (0.84–1.63), O_3 0.58 (0.33–0.83); overall pollutant change corresponded to a 6.6 percentage-point reduction in predicted risk. Two real limits: the outcome is a validated *prediction*, not observed incidence, and treatment assignment is a municipal policy target, so residual regional confounding is structural rather than adjustable. O_3 rose in both arms.

doi: 10.1016/j.lanplh.2026.101513

Reproductive & developmental 1 record

Gao et al. 2026 Environ Sci Technol

Urinary biomarkers of airborne organics in pregnancy and small-for-gestational-age risk

TIMFEM prospective cohort, n=1,425 mother–infant pairs, 20 urinary biomarkers of airborne organic pollutants by LC-MS/MS (p-phenylenediamines, nitro-PAHs, phenylguanidines, benzothiazoles, cotinine), 18 detected in >40% of participants, with leave-one-out in vitro screening in human trophoblast cells at exposure-guided doses. Each ln-unit increase in **6PPD was associated with 27% higher SGA risk and 6PPD-Q with 23%**; mixture models added TPG, cotinine and 1-aminopyrene. The mechanism the cell work supports is metabolic rather than cytotoxic — reduced pyruvate entry into the TCA cycle and impaired isocitrate-to-α-ketoglutarate conversion at doses that barely touched proliferation. No confidence intervals are given for the epidemiological estimates in the abstract, so they are not plotted here. Relevance: these are tyre-and-road and traffic-associated organics that no PM mass channel resolves.

doi: 10.1021/acs.est.6c04451

Respiratory & allergic 1 record

Gómez-Olarte et al. 2026 Environ Pollut

Early-life air pollution, respiratory outcomes and stimulated cytokine response

German birth cohort, n=606 one-year-olds for bronchitis and wheezing, cytokine subcohort n=508 (LPS-stimulated TNF-α, MCP-1, IL-8, IL-6, IL-10), residential $PM_{2.5}/PM_{10}/NO_2$ per IQR of 2.0/2.8/9.1 $\mu g\,m^{-3}$, with BKMR for the mixture and stratification by sex and temperature. In boys, $PM_{2.5}$ gave **aOR 1.50 (1.02–2.23)** for bronchitis and **+23.8% (11.2–37.9)** IL-8; in girls, **+31.9% (18.6–46.7)** IL-8, +12.8% (3.9–22.6) TNF-α and +35.7% (14.4–61.1) IL-10. The mixture association with IL-8 and TNF-α was non-linear and appeared only above the 60th percentile of all pollutants. Small cohort, and the sex split doubles the tests; the consistent direction across five cytokines is what carries it. $PM_{2.5}$, not NO_2, is the primary predictor.

doi: 10.1016/j.envpol.2026.129079

Mechanistic toxicology 2 records

Meldrum & Leonard 2026 Front Immunol

Diesel particle priming reprograms the airway response to a subsequent allergen

Murine model with deliberate temporal separation between diesel exhaust particle exposure and house dust mite challenge — so neither co-exposure nor prior sensitisation — read out with compartment-resolved bulk transcriptomics (lung tissue vs airway lumen) plus targeted single-cell profiling. DEP priming amplified leukocyte recruitment on subsequent HDM challenge and did so **neutrophilically, with no eosinophil response**. DEP alone drove NF-κB programmes; HDM drove NF-κB and interferon; the primed response was qualitatively different from either, with elevated ISG expression and expanded neutrophil subpopulations concentrated in the luminal compartment. The design is the contribution: it isolates a memory-like effect of a particle exposure that has already cleared. Murine, single particle type, no dose–response.

doi: 10.3389/fimmu.2026.1869644

Arora et al. 2026 J Environ Sci (China)

Oxidative potential of fresh versus ozone-aged $PM_{2.5}$ across Chinese sources

A commentary, not primary data — it summarises Ma et al. (`10.1016/j.jes.2024.04.023`), which measured DTT-assay oxidative potential on fresh and O_3-aged $PM_{2.5}$ from multiple source types. **Biomass-burning particles carried up to 35× the OP of suburban $PM_{2.5}$**, driven by water-soluble organics and transition metals, and O_3 aging generally *lowered* OP while producing complex chemical transformation. Counted and listed here because the source-dependence claim is the one this watch tracks, but no number in it is new and the aging direction runs opposite to the usual assumption that atmospheric processing increases toxicity. Treat as a pointer to the 2024 primary paper.

doi: 10.1016/j.jes.2026.05.042

Occupational & indoor 2 records

Yang et al. 2026 Atmosphere

Ventilation optimisation for a 300 m welding workshop with distributed dust sources

CFD of airflow and welding-fume transport in a 300 × 28 × 21 m hall, varying exhaust-to-supply ratio and side-exhaust-port height, and contrasting low-density aluminium-alloy fume with high-density carbon-steel fume. Two air-velocity peaks appear at 0–2 m and 8–12 m above floor; the **top exhaust port outperforms the side port** throughout, higher ESR accelerates upward migration and suppresses lateral dispersion, and a badly set side-port height produces either short-circuiting or excess lateral spread. Optimal side-port height **5 m for aluminium, 6 m for carbon steel**. Simulation only — no measured personal exposure, and fume is treated as a passive density-differentiated tracer rather than a coagulating aerosol.

doi: 10.3390/atmos17090843

Jaiswal et al. 2026 Research Square (preprint)

Nature-based indoor filtration in an EDM gear-manufacturing shop

Single-site five-month before/after intervention in Gurugram, India. Baseline was severe: **$PM_{2.5}$ 330 μg m^{-3} and CO 88 ppm**. Plant-based filter units plus a redesigned vent/exhaust layout, sited using airflow visualisation, cut **CO by 90% and $PM_{2.5}$ by 42% on average** — leaving $PM_{2.5}$ around 190 μg m^{-3}, which the authors say plainly is still not a healthy range. No control period, no controlled comparison against a conventional HEPA unit, and the CO and PM reductions differ by a factor the paper does not explain mechanistically. Preprint, not peer reviewed. Listed for the occupational baseline number, which is the part that is hard to obtain.

doi: 10.21203/rs.3.rs-10511730/v1

Sensing, forecasting & instrumentation 5 records

Raso et al. 2026 Atmosphere

Temporal resolution and the characterisation of episodic pollution events

Two consecutive New Year's Eve firework campaigns with certified reference instrumentation co-deployed with calibrated low-cost sensors, comparing daily, hourly and minute aggregation of PM_{10}, NO_2, CO and O_3. Regulatory averaging detects the episode and misses its structure: minute PM_{10} reached **215 μg m^{-3}**, a **peak amplification factor of 2.90–2.93** over the hourly value. The direct implication for distributed sensing is that the case for a dense network is not redundancy against reference-grade accuracy but access to a time scale the reference framework discards — and that acute-exposure metrics built on hourly data are systematically low for episodic sources. Two events at one site; the PAF is a description, not a transferable constant.

doi: 10.3390/atmos17090860

Mohd Nadzir et al. 2026 Atmosphere

Minute-scale trace gas and PM variability in the Antarctic Peninsula from a low-cost node

A single low-cost multi-sensor package, May–June 2023, **58,652 valid observations** of NO_2, O_3, CO, SO_2, $PM_{2.5}$ and PM_{10} (TVOC in May only). Medians: $PM_{2.5}$ **6.29** and PM_{10} **7.29 μg m^{-3}**, O_3 38.17 ppb, NO_2 11.05 ppb, SO_2 0.01 ppb. The **$PM_{2.5}$/PM_{10} ratio of 0.96** indicates an almost entirely fine-mode aerosol, consistent with combustion rather than sea salt or crustal dust. Percentile-based event detection found **142 single- and 30 multi-pollutant enhancements**. The TVOC channel was 98% zeros and was rightly demoted. What is missing is any co-location or drift assessment — at these concentrations a few μg m^{-3} of baseline offset would change every conclusion, and an uncalibrated nephelometric PM channel in high humidity is exactly where that offset lives.

doi: 10.3390/atmos17090855

Alraddadi et al. 2026 Atmosphere

Explainable one-hour-ahead AQI forecasting across eight Saudi cities

Attention-based bidirectional recurrent networks against a persistence baseline in eight climatically distinct cities. The deep models **matched persistence on RMSE and R^2** (cross-city mean R^2 ≈0.90) and beat it on MAE by roughly a fifth, with the margin largest in the most variable cities. Event-based evaluation flagged **84–91% of high-AQI exceedances**. Surface meteorology added little on average and its value was city- and architecture-specific. SHAP and LIME put particulate matter as the dominant driver everywhere. The persistence comparison is the part worth copying: at one-hour horizon hourly AQI is so autocorrelated that an unbenchmarked R^2 of 0.90 means nothing, and most papers in this genre do not report the baseline.

doi: 10.3390/atmos17090858

Chen C et al. 2026 J Hazard Mater

Amine-containing particles from fireworks by single-particle mass spectrometry

2023 Spring Festival field campaign with single-particle aerosol mass spectrometry plus source samples. Amine-containing particles rose **96.2%** during the firework period, peak size shifted **0.42** → **0.44** μ**m**, and internal mixing with sulfate increased; fireworks raised metal and elemental-organic-carbon classes while depressing biomass-burning and secondary classes. **N-methylaniline** is proposed as a tracer separating fireworks from firecrackers. A multi-model comparison identifies modified aerosol acidity, not liquid water content, as the dominant control on gas–particle partitioning of amines, with ALWC showing a non-linear threshold. One festival, one city; the tracer claim needs a second site before it is usable.

doi: 10.1016/j.jhazmat.2026.143446

Hong et al. 2026 Indoor Air

Predicting fungal and bacterial bioaerosol concentrations across building types

9,377 indoor air samples across 27 multiuse facility types in South Korea, modelled with a generalised additive mixed model to capture non-linear temperature and humidity effects, then a random forest on the resulting sensitivity classes. Fungi respond mainly to relative humidity (16 facility types), bacteria mainly to temperature (7); peak microbial levels occur at **55–60% RH**, supporting 60% as the practical dampness threshold. Stratifying by meteorological sensitivity before prediction raised F1 from 0.53 to **0.65** (**bacteria**) **and 0.75** (**mould**). Culture-based counts, so the non-viable fraction is invisible, and the facility-type classes are administrative rather than physical. Directly relevant to anyone adding T/RH channels to an indoor node and expecting them to carry bioaerosol information.

doi: 10.1155/ina/6724151

Exposure assessment & modelling — 16 records

Wang T et al. 2026 J Environ Sci (China)

IAQMS-Industry: 100 m $PM_{2.5}$ inside industrial parks

Couples the regional NAQPMS to a city-scale chemical transport model with explicit point-source locations and Gaussian plume dispersion, resolving $PM_{2.5}$ at **100 m** against a 3 km regional baseline, evaluated at Beijing Yizhuang and Tangshan. Normalised mean bias improves from -16.9%–-7.7% to **3.1–6.2%**, and industrial point sources account for **22.9–26.4%** of park $PM_{2.5}$ versus 1.6–13.7% in the regional model. The mechanism is straightforward — coarse grids smooth the concentration gradient and over-disperse — but the size of the attribution error is the useful number. This is the exact error a dense in-park sensor network would be deployed to measure, and the paper supplies the modelled expectation to test against.

doi: 10.1016/j.jes.2025.12.020

Boyle et al. 2026 Cancer Epidemiol Biomarkers Prev

Cancer-specific air-pollution exposome metrics for the continental US

Combines CACES land-use-regression criteria-pollutant surfaces with the RSEI industrial-release model across US census tracts, 2005–2020, weighting each pollutant by meta-analytic breast- and lung-cancer risk estimates to produce two composite metrics (APBC, APLC). Both are strongly right-skewed — some tracts an order of magnitude above the mean — peak in large cities and along transport corridors, differ from RSEI notably in Los Angeles and New York, vary by race, ethnicity and housing, and cluster spatially. No cancer outcome is analysed here: this is metric construction, and the meta-analytic weights import the heterogeneity of their source studies into a tract-level surface that then reads as precise. Useful as a template for building a risk-weighted composite from a network, with that caveat attached.

doi: 10.1158/1055-9965.EPI-26-0493

Arpornwichanop & Chuersuwan 2026 Environ Geochem Health

Life-stage-resolved inhalation exposure to $PM_{2.5}$-bound elements, Greater Bangkok

Secondary-data screening at one fixed site (selected 24 h filter days, Dec 2020–Sep 2021), combining seasonally weighted outdoor elemental concentrations with indoor penetration, time-activity, respiratory-deposition and simulated-lung-fluid solubility assumptions across pediatric life stages. Concentration and deposited-intake profiles rank **Al, Fe, B, Zn** highest; the solubility- and toxicity-adjusted comparative priority quotient ranks **As, Cd, Sb**. Modelled lifetime cancer risk spans $\mathbf{1.11\times10^{-6}}$ **to** $\mathbf{2.78\times10^{-5}}$ across 0–100% Cr(VI) assumptions — an order of magnitude driven entirely by a speciation parameter nobody measured. The authors state explicitly that this is within-model prioritisation, not measured child exposure. That honesty is unusual and worth noting.

doi: 10.1007/s10653-026-03455-7

Xiao S et al. 2026 Environ Geochem Health

PAHs and oxygenated PAHs in Xi'an $PM_{2.5}$ under the clean air action

Winter $PM_{2.5}$-bound parent PAHs and OPAHs in Xi'an, a Fenwei Plain basin city, with PMF source apportionment and toxic-equivalent concentrations. Wintertime PAHs are lower than historical levels under the clean-air actions; biomass-burning-related mixed combustion is the largest PMF factor. During haze episodes both classes rise, but the increase is **statistically significant for OPAHs and not for parent PAHs**, and **OPAHs — BcdPQ in particular — dominate the toxic-equivalent burden**. Regional pollution is in-basin accumulation plus short-range transport from northern Shaanxi industry, modulated by synoptic pressure pattern. The finding that matters: a control programme that lowers parent PAHs need not lower the oxygenated derivatives that carry the toxicity, and routine monitoring measures the former.

doi: 10.1007/s10653-026-03449-5

Gürler Akyüz & Akyüz 2026 J Environ Sci Health A

Size-resolved elemental composition in a coastal urban-industrial region

Seasonal $PM_{2.5}$ and $PM_{2.5-10}$ elemental composition at Zonguldak, Türkiye, where coal mining, coal combustion, iron-steel with coke production, maritime fuel oil, traffic and marine/crustal sources overlap. Heating-period $PM_{2.5}$ carries elevated **As, Sb, Cd, Pb, Se, V, Ni, Cr**; non-heating $PM_{2.5-10}$ is crustal, marine and resuspended dust with a persistent industrial residue. Enrichment factors against Ti show strong anthropogenic enrichment for combustion and metallurgy and near-natural levels for crustal and marine elements; PCA separates coal combustion, iron-steel, fuel-oil/shipping, non-exhaust traffic, crustal dust, marine aerosol and coal fly ash, and discriminant analysis confirms the season/size separation. Conventional in method, valuable as a composition baseline for a source mixture that is rarely characterised together.

doi: 10.1080/10934529.2026.2724190

Guo & Sudo 2026 Atmosphere

Global online PAH simulation in the CHASER chemistry-climate model

First global online simulation of naphthalene, phenanthrene, pyrene and benzo[a]pyrene in CHASER V4.0, coupling soot adsorption $K_{SA}(T)$ with organic-matter absorption $K_{OA}(T)$ under an extended Dachs–Eisenreich dual-mode scheme. Evaluated at urban, background and Arctic stations: pooled spatial R **0.78** on a $\log_{10}$ scale, **FAC2 80%** for BaP excluding sub-grid hotspots, seasonal R 0.62–0.93. The K_{SA} term fixes the long-standing under-prediction of particle-bound fraction at cold high latitudes, raising polar ϕ_{BaP} by up to ∼0.6, and elevated elemental carbon extends BaP lifetime to **1.72 d** against 0.57–0.72 d for lighter congeners, sustaining transoceanic outflow. The dependence on modelled EC is also the weak point: the partitioning result is only as good as the soot field underneath it.

doi: 10.3390/atmos17090861

Klebach et al. 2026 Atmos Chem Phys

MSA and sulfuric acid in tropical Indo-Pacific aerosol from aircraft

CAFE-Pacific, HALO aircraft, January–February 2024, nitrate chemical-ionisation mass spectrometry of methanesulfonic and sulfuric acid. Marine boundary layer concentrations reach **4×10^7 cm^{-3} MSA and 6×10^7 cm^{-3} SA**; the MSA/SA ratio increases with altitude in the lower free troposphere, consistent with temperature-dependent DMS oxidation. Two flights show marine deep convection lofting DMS to the upper troposphere with particle formation and growth after **10–20 h of OH exposure**, matching the kinetically modelled DMS lifetime. MSA frequently exceeds SA, which the current generation of new-particle-formation schemes does not anticipate. Aircraft inlets heat and evaporate acidic particles, so the gas/particle split at altitude is instrumental as much as atmospheric — the authors are explicit about this.

doi: 10.5194/acp-26-12355-2026

Papetta et al. 2026 Atmos Chem Phys

Source-dependent optical and mineral signatures of Mediterranean dust outbreaks

Four major 2021–2022 dust outbreaks from the Eastern, Western and Central Sahara and the Middle East, tracked across **24 AERONET stations** with IASI and MODIS MIDAS dust optical depth and HYSPLIT back-trajectories. Saharan events were coarse and scattering; the Middle East event carried **finer, more absorbing particles**, likely anthropogenically influenced. DOD-to-AOD ratio separates them — only one East-Central Saharan event stayed above 0.8 (near-pure dust), while the Middle East event had the lowest ratios, indicating mixing with anthropogenic or marine aerosol. A Cyprus in-situ case study with elemental and absorption measurements anchors the remote-sensing inference. For network work the transferable idea is the DOD/AOD purity screen: it is a cheap way to decide whether a coarse-mode excursion is genuinely dust before attributing it.

doi: 10.5194/acp-26-12395-2026

Liu S et al. 2026 J Environ Sci (China)

Terrain blocking and monsoon coupling in North China Plain $PM_{2.5}$ and O_3

Quantifies mountain-blocking effects on pollutant dispersion in the semi-closed North China Plain now that anthropogenic emissions have fallen far enough for terrain and meteorology to dominate the residual pattern. Within 120 km buffers of the major ranges, concentrations rise with altitude and then fall sharply above a threshold; the **effective blocking altitude is 22–929 m for winter $PM_{2.5}$ and 19–1060 m for summer O_3**. Concentrations rank plains > platforms > hills > mountains, and winter $PM_{2.5}$ and summer O_3 high-value zones aggregate in *opposite* north–south directions. Descriptive rather than causal — emissions and terrain co-vary strongly in the NCP — but the altitude thresholds are a concrete siting constraint for any regional network.

doi: 10.1016/j.jes.2026.01.013

Stergiou et al. 2026 Atmos Pollut Res

Spatiotemporal graph neural network bias correction for CMAQ over CONUS

Ships on metadata only. Crossref carries no abstract for this DOI and the record is not in PubMed, Europe PMC or Semantic Scholar as of this run, so nothing beyond title, authors and venue can be attributed to it. Counted and listed; no numbers claimed. Flagged for the abstract retry list — graph-structured bias correction of a regulatory CTM against monitor networks is directly on this watch's topic and should be summarised once the abstract deposits.

doi: 10.1016/j.apr.2026.103191

Xu L et al. 2026 J Environ Sci (China)

Marine aging compacts soot and thickens its coating

TEM single-particle analysis of soot collected over the Bohai Sea and the Northern and Southern Yellow Sea after outflow from East Asia. **Over 98%** of soot-containing particles are coated, mean mixing-state index **0.83**. Fractal dimension rises along the transport path — 1.84 ± 0.05 (N. Yellow Sea), 1.90 ± 0.08 (Bohai), 1.96 ± 0.07 (S. Yellow Sea) — indicating progressive structural compaction, while D_p/D_{core} of 3.9–5.3 exceeds the 3.54 typical of continental regional transport. Compact, uniformly coated, thickly shelled soot absorbs substantially more than the lacy fresh particle that optical models assume. Any absorption-based black-carbon instrument calibrated on fresh soot will misreport aged marine air, and this paper quantifies by how much the morphology has moved.

doi: 10.1016/j.jes.2026.01.047

Zhang N et al. 2026 J Environ Sci (China)

Aging, composition and light absorption of nitrogen-containing organics in PM_1

59 Shanghai PM_1 samples analysed by UHPLC-Orbitrap MS. Nitrogen-containing organics are **31% of detected species in ESI− and 64% in ESI+**. Aging reduces molecular diversity: CHON− compounds fall in both number and mass fraction with aging degree, and carboxylic-rich alicyclic molecules drop from **37.7% to 21.2%** of the CHON− contribution. Mass absorption efficiency at 365 nm declines with aging alongside the Bep/(Bep+Bap) ratio and relative humidity, pointing to aqueous-phase oxidation as the dominant mechanism, with CHON− and CHN+ identified as the principal chromophores. Semi-quantitative by design — ESI response factors are compound-specific and uncalibrated — so the fractions are ranks, not concentrations.

doi: 10.1016/j.jes.2025.12.003

Xu X et al. 2026 J Environ Sci (China)

Ammonia enhances isoprene SOA through gas-phase cluster formation

Smog-chamber series on isoprene photooxidation by OH with and without NH_3, with nitrate-CIMS for the gas phase and HR-TOF-AMS for the particle phase. NH_3 increased SOA mass and accelerated isoprene oxidation. The mechanism is not the expected particle-phase acid–base reaction: CIMS shows NH_3 reacting homogeneously in the *gas* phase with organic acids and low-volatility oxygenated organic molecules to form extremely and ultra-low-volatility **NH_3–OOM clusters**, which then nucleate and condense. Quantum chemistry confirms spontaneous hydrogen-bonded association with RCOOH, R-OOH and R-OH groups. **Cluster formation accounts for 78% of the SOA enhancement.** Chamber conditions with elevated NH_3; the relevance is to agricultural and livestock regions where NH_3 and biogenic VOC co-occur.

doi: 10.1016/j.jes.2026.01.041

Kong J et al. 2026 J Environ Sci (China)

Triplet-state brown carbon as an endogenous oxidant

Photochemistry of 1,4-naphthoquinone as a brown-carbon proxy in deoxygenated aqueous systems under 355 nm irradiation, resolving two long-lived $^3(n,\pi^*)$ triplets (^{3}NQ* and ^{3}OH-NQ*). ^{3}NQ* abstracts hydrogen from organic substrates **without any external oxidant**, with a rate hierarchy set by bond dissociation energy and one-electron oxidation potential: alkanes $\sim 10^4$, oxygenated alkanes $\sim 10^5$, phenols $\sim 10^6$ $M^{-1}\,s^{-1}$. Self-photoconversion regenerates a second triplet, amplifying the oxidative cascade. The claim is that ^{3}BrC* is the dominant phenol sink in BrC-rich haze under acidic, O_2-limited conditions — which is exactly the condition inside a haze particle and exactly the condition current models do not represent. Proxy compound, deoxygenated system; the rate constants are the transferable output.

doi: 10.1016/j.jes.2025.10.037

Wang J et al. 2026 J Environ Sci (China)

Water-soluble aerosol matter photosensitises guaiacol into aqueous SOA

Mechanism and kinetics of aqSOA formation from guaiacol mediated by photosensitive water-soluble inorganic and organic matter extracted from real aerosol. Photolysis of the extract generates both reactive oxygen species and triplet WSOM; **^{3}WSOM* is the predominant initiator** of guaiacol transformation, with steady-state concentration **$5.35 \pm 0.12 \times 10^{-13}$ mol/L**, generation rate 3.43×10^{-7} mol $L^{-1}\,s^{-1}$, quantum yield **4.86%** and a rate constant with guaiacol above 10^9 L $mol^{-1}\,s^{-1}$. ROS then convert soluble intermediates into the insoluble ethers and low-volatility phenols that constitute the aqSOA. Bulk aqueous extracts are not cloud or particle water — ionic strength differs by orders of magnitude — which is the standing limitation of this whole method.

doi: 10.1016/j.jes.2025.11.050

Solaiman et al. 2026 Environ Sci Atmos

Nighttime NO_3 oxidation of myrtenal: SOA yields and HOMs

Chamber study of secondary organic aerosol from the gas-phase oxidation of myrtenal, a biogenic aldehyde, by NO_3 radicals, reporting SOA yields, chemical composition and highly oxygenated molecules. The Crossref deposit truncates the abstract mid-sentence and no full record was retrievable from Semantic Scholar on this run, so the yields and HOM distributions cannot be quoted. Counted and listed; the nighttime NO_3 channel for oxygenated biogenics is under-represented in this watch's corpus and the record is worth revisiting once the full abstract deposits.

doi: 10.1039/d6ea00092d

Burden, policy & mitigation — 3 records

Zhao N et al. 2026 J Hazard Mater

Attainment pressure under China's revised GB 3095-2026 standard in Shandong

Daily observations of six criteria pollutants across 16 Shandong cities, 2015–2025, condensed into a relative attainment pressure index combining concentration gap and additional non-attainment days, with random forest, SHAP and monotonicity-constrained XGBoost on socioeconomic predictors. **$PM_{2.5}$, PM_{10}, SO_2, NO_2 and CO fell 41.0–79.6%; O_3 did not move, staying at 166–191 $\mu g\,m^{-3}$.** Under the transition-stage limits a western high-pressure belt emerges — Heze 98.7, Dezhou 91.3, Liaocheng 89.1. Civilian vehicle ownership and secondary-industry share are the leading predictors for particulates and NO_2; the socioeconomic set explains $PM_{2.5}$ far better than O_3. Even under the deep-transition pathway **eight of sixteen cities remain above 30 $\mu g\,m^{-3}$ in 2030**. One province, and the scenario arithmetic assumes the historical predictor–concentration relation holds forward.

doi: 10.1016/j.jhazmat.2026.143460

Li Y et al. 2026 Environ Sci Technol

Multidimensional inequality of CO_2 and air-pollutant footprints in China

Consumption-based per-capita footprint accounting for CO_2 and air pollutants across 2020–2060 under the dual-carbon policy. The top 10% carry **28.5 t CO_2 and 50.6 kg NO_x** per capita against **1.2 t and 4.7 kg** for the bottom 50%. The structural finding is that **over 90% of footprint inequality is intraprovincial**, not between provinces, and the gap widens to 2060 — which means province-level policy instruments are aimed at the wrong axis. Composition differs by stratum: the top 1% footprint is investment-driven (96% of SO_2), the bottom 50% consumption-driven (88% of NH_3). Input–output footprint accounting carries well-known allocation sensitivity, and the 2060 horizon is a scenario, not a projection.

doi: 10.1021/acs.est.6c04314

Shi et al. 2026 Environ Sci Technol

Carbon–air pollution cobenefits and emission leakage in China's steel industry

Unit-level emission database covering **2,866 enterprises and ~4,030 units**, relating regional energy-industrial configuration to mitigation priorities. Steel CO_2 peaked at 1.99 Gt in 2019 while SO_2 and PM fell **more than 80%** over 2011–2022 — the decoupling this watch keeps seeing. Over 2023–2050 the priorities diverge: production restraint gives the largest cumulative CO_2 reduction (22.1%) while **obsolete-capacity retirement dominates air-pollutant reduction (22.3%)**, so a carbon-optimal pathway is not a PM-optimal one. Regionally differentiated pathways induce **spatial emission leakage**, stronger for CO_2 (17.39% in 2030, falling to 2.38% by 2050) than for pollutants, and national mean pollutant concentrations fall only 5.4–12.3% by 2030 relative to 2022. Sector-specific and inventory-based.

doi: 10.1021/acs.est.6c08461

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Barbalat et al.	Association of heat, cold, and ambient pollution in the first 1000 days of life with autism spectrum disorder risk indicators in two-year-olds: findings from the ELFE national birth cohort	*Environ Sci Technol*	Prospective birth cohort + DLNM	Weekly mean $PM_{2.5}$, PM_{10}, NO2 from 1 km daily outdoor predictions (200 m for NO2/ temperature in large urban areas)	France
Kuznetsov et al.	Particulate matter air pollution and life satisfaction: links across and within twin pairs	*Environ Res*	Twin-pair longitudinal comparison	Residential $PM_{2.5}$ and PM_{10}, repeated across six waves	Germany
Cardiovascular & metabolic					
Sun D et al.	Exploring China's Clean Air Act and associated cardiovascular disease risk: a prospective, quasi-experimental, and causal inference modelling study	*Lancet Planet Health*	Quasi-experimental (difference-in-differences)	Long-term $PM_{2.5}$, PM_{10} and O3; municipal PM-reduction target as the intervention assignment	China
Reproductive & developmental					
Gao et al.	Exposure biomarkers of airborne organic pollutants in pregnant women and developmental risks: evidence from epidemiology to in vitro assessment	*Environ Sci Technol*	Prospective cohort + in vitro screening	20 urinary biomarkers of airborne organics (6PPD, 6PPD-Q, nitro-PAHs, phenylguanidines, benzothiazoles, cotinine) by LC-MS/MS	China
Respiratory & allergic					
Gomez-Olarte et al.	Early-life exposure to ambient air pollution and respiratory outcomes: mixture and sex-stratified associations with inflammatory cytokine responses	*Environ Pollut*	Prospective birth cohort	Residential annual $PM_{2.5}$, PM_{10}, NO2; per IQR 2.0 / 2.8 / 9.1 ug m-3	Germany
Mechanistic toxicology					
Arora et al.	Oxidative potential of fresh vs. O3-aged $PM_{2.5}$ across urban and rural sources in China	*J Environ Sci* (*China*)	Commentary / methods letter	$PM_{2.5}$ oxidative potential by DTT assay, fresh vs ozone-aged, source-resolved	China
Meldrum & Leonard	Pollutant particle priming amplifies airway compartment specific neutrophil and macrophage inflammatory programs following house dust mite acute exposure	*Front Immunol*	Murine inhalation exposure	Diesel exhaust particles, temporally separated priming dose before allergen challenge	United Kingdom
Occupational & indoor					
Jaiswal et al.	Using nature-based solutions in mitigating indoor air quality: a case study of EDM emission from a gear manufacturing unit, India	*Research Square* (*preprint*)	Trial / intervention	Indoor $PM_{2.5}$ and CO before and after five months of NBS filtration plus ventilation redesign	India
Yang et al.	Optimization of ventilation systems in large welding workshops with multiple dust sources: a case study of a 300 m long welding workshop	*Atmosphere*	Chemical transport modelling	Welding fume transport by CFD; low-density aluminium vs high-density carbon-steel fume	China
Sensing, forecasting & instrumentation					
Alraddadi et al.	Explainable one-hour-ahead air quality index forecasting across Saudi cities using pollutant and meteorological data	*Atmosphere*	Algorithm development	Hourly AQI (PM-dominated) with and without surface meteorological predictors	Saudi Arabia

First author	Focus	Journal	Design	Metric	Region
Chen C et al.	Characteristics and gas-particle partitioning of amine-containing particles from firework emissions: new insights from machine learning and single-particle mass spectrometry	*J Hazard Mater*	Real-time sensor campaign	Single-particle aerosol mass spectrometry of amine-containing particles; modified aerosol acidity Rra′	China
Hong et al.	Predicting fungal and bacterial bioaerosol concentrations across diverse building types: a framework integrating generalized additive models and machine learning	*Indoor Air*	Cross-sectional + prediction model	Culturable bacterial and fungal bioaerosol concentrations vs indoor temperature and relative humidity	South Korea
Mohd Nadzir et al.	Short-term variability of trace gases and particulate matter in the Antarctic Peninsula atmosphere using a low-cost air sensor	*Atmosphere*	Real-time sensor campaign	Minute-scale NO2, O3, CO, SO2, $PM_{2.5}$, PM_{10} (plus TVOC in May) from a single low-cost multi-sensor node	Antarctica
Raso et al.	Influence of temporal resolution on the characterization of episodic air pollution events: experimental evidence from New Year's Eve fireworks	*Atmosphere*	Real-time sensor campaign	PM_{10}, NO2, CO, O3 at minute, hourly and daily resolution; certified reference instruments alongside calibrated low-cost sensors	Italy
Exposure assessment & modelling					
Arpornwichanop & Chuersuwan	Life-stage-resolved screening of inhalation exposure to $PM_{2.5}$-bound elements in Nonthaburi, Greater Bangkok, Thailand	*Environ Geochem Health*	Exposure model (secondary data)	$PM_{2.5}$-bound elements; respiratory-deposited intake adjusted by simulated-lung-fluid solubility	Thailand
Boyle et al.	Creating outdoor air pollution exposome metrics for assessing breast and lung cancer risks in the United States	*Cancer Epidemiol Biomarkers Prev*	Metric development + retrospective application	Census-tract criteria-pollutant concentrations (CACES land-use regression) plus RSEI industrial-release model, weighted by cancer-specific meta-analytic risk estimates	United States
Guo & Sudo	Global simulation of polycyclic aromatic hydrocarbons (PAHs) with the CHASER chemistry-climate model	*Atmosphere*	Chemical transport modelling	Gas-particle partitioning of NAP, PHE, PYR and BaP under a coupled soot-adsorption / organic-matter-absorption scheme	Global
Gurler Akyuz & Akyuz	Seasonal and size-resolved elemental composition and source apportionment of atmospheric particulate matter in a coastal urban-industrial region of Zonguldak, Turkiye	*J Environ Sci Health A*	Field measurement	$PM_{2.5}$ and $PM_{2.5}$-10 elemental composition; enrichment factor, PCA and discriminant analysis	Turkiye
Klebach et al.	Aircraft observations suggest an important contribution of methanesulfonic and sulfuric acids to tropical Indo-Pacific aerosol	*Atmos Chem Phys*	Aircraft measurement campaign	Gas- and particle-phase MSA and H2SO4 by nitrate chemical-ionisation mass spectrometry	Indo-Pacific
Kong J et al.	Investigating the photosensitized chemistry of proxies for tropospheric brown carbon	*J Environ Sci (China)*	Bench / device experiment	Excited triplet-state brown carbon proxy (1,4-naphthoquinone); aqueous-phase oxidation kinetics	Laboratory

First author	Focus	Journal	Design	Metric	Region
Liu S et al.	Quantifying the combined effects of semi-closed terrain and monsoon on $PM_{2.5}$ and O3 pollution aggregation in the North China Plain	*J Environ Sci (China)*	Spatial analysis / GIS	Gridded $PM_{2.5}$ and O3 stratified by landform and altitude; Terrain-Wind Close Index	China
Papetta et al.	Source-dependent optical and mineral signatures of dust outbreaks over the Mediterranean	*Atmos Chem Phys*	Multi-platform observation synthesis	Dust optical depth, AOD, single-scattering albedo and asymmetry factor; AERONET + IASI + MODIS MIDAS + HYSPLIT	Mediterranean
Solaiman et al.	Secondary organic aerosol formation from the gas-phase oxidation of myrtenal by NO3 radicals: SOA yields, chemical composition, and highly oxygenated molecules	*Environ Sci Atmos*	Smog chamber experiment	SOA mass yields and HOM composition from a biogenic aldehyde oxidised by NO3 at night	Laboratory
Stergiou et al.	A spatiotemporal graph neural network framework for retrospective bias correction of CMAQ air quality simulations over the continental United States	*Atmos Pollut Res*	Algorithm development	CMAQ-simulated criteria pollutants including $PM_{2.5}$, monitor-anchored bias field	United States
Wang J et al.	Water-soluble matter mediated phototransformation of guaiacol leading to formation of aqueous secondary organic aerosol: mechanism and kinetics	*J Environ Sci (China)*	Bench / device experiment	Aqueous SOA from guaiacol photosensitised by water-soluble aerosol extracts; ROS and triplet WSOM	Laboratory
Wang T et al.	An integrated multiscale air quality modelling framework for industrial park pollution: linking local emissions to regional transport	*J Environ Sci (China)*	Chemical transport modelling	$PM_{2.5}$ at 100 m resolution from NAQPMS coupled to a city-scale CTM with explicit Gaussian point-source plumes	China
Xiao S et al.	Pollution characteristics and health risks of PAHs and OPAHs in $PM_{2.5}$ of an inland basin city in China under the influence of the clean air action plan	*Environ Geochem Health*	Source-apportionment method	$PM_{2.5}$-bound parent PAHs and oxygenated PAHs, PMF source resolution, toxic-equivalent concentration	China
Xu L et al.	Marine air promotes structural compaction and coating growth of soot aerosols after long-range transport from East Asia	*J Environ Sci (China)*	Field measurement	Single-particle soot morphology and mixing state by TEM; fractal dimension and Dp/Dcore	China
Xu X et al.	Enhancement of secondary organic aerosol formation from isoprene photooxidation by ammonia	*J Environ Sci (China)*	Smog chamber experiment	Isoprene SOA mass and composition; nitrate-CIMS gas phase plus HR-TOF-AMS particle phase	Laboratory
Zhang N et al.	Effects of aerosol aging on composition and light-absorbance of nitrogen-containing organic compounds: evidences from ultra-high-resolution mass spectrometry analysis	*J Environ Sci (China)*	Field measurement	PM1 nitrogen-containing organics by UHPLC-Orbitrap MS; mass absorption efficiency at 365 nm	China
Burden, policy & mitigation					
Li Y et al.	Multidimensional inequality of CO2 and air pollutant footprints in China under dual-carbon policy	*Environ Sci Technol*	Modelling / inventory	Per-capita consumption-based footprints of CO2, SO2, NOx, NH3 and primary PM	China

First author	Focus	Journal	Design	Metric	Region
Shi et al.	Coupled energy-industrial transitions shape heterogeneous carbon-air pollution cobenefits and spatial emission leakage in China's steel industry	*Environ Sci Technol*	Modelling / inventory	Unit-level SO2 and primary PM emissions alongside CO2, with modelled ambient response	China
Zhao N et al.	Uneven attainment pressure under China's tightened ambient air quality standard: socioeconomic predictors and differentiated control pathways for $PM_{2.5}$ and O3 in Shandong	*J Hazard Mater*	Spatial regression (GWR)	Daily $PM_{2.5}$, PM_{10}, SO2, NO2, CO and O3, 2015-2025; relative attainment pressure index against GB 3095-2026	China

Provenance and method

Entry window **1 September 2026**, single day, opened against a `last_entry_date` of 31 August so no day is skipped. Sources queried on 3 September 2026 (local 20:5x CDT, so under this pipeline's 22:00 rule the current day is not yet buildable and 1–2 September are the owed issues). **PubMed** E-utilities on `[EDAT]` 2026/09/01: 26 hits on the health axis and 5 on the instrumentation axis, run as separate queries. **Europe PMC** `CREATION_DATE`: 11 hits. **Crossref** by-ISSN, windowed on `created`: 97 deposits across the tracked atmospheric and exposure journals — the channel that supplies *Atmos. Chem. Phys.*, *Atmosphere* and *Environ. Sci. Atmos.*, none of which PubMed indexes. **Consensus** on the sensor-calibration axis: 3 hits, **all 3 already in the store**. **OpenAlex** returned HTTP 403 (plan-gated date filter) and **arXiv** HTTP 429 on repeated attempts; both legs are recorded as failed, not as empty. After collapsing cross-source duplicates by DOI and normalised title, 135 unique records were considered; **4 were already shipped** (all on 31 August, caught on the PubMed identifier leg), 33 were screened in scope and 98 were screened out as off-topic — predominantly aquatic and soil chemistry, microbiology, meteorology and non-particulate building-science deposits carried in by the by-ISSN sweep. Subtopic, design, particle-metric and geography labels are single-label by dominant contribution and are a judgement, not a controlled vocabulary. Effect estimates are transcribed verbatim from abstracts. Two records (`10.1016/j.apr.2026.103191`, `10.1039/d6ea00092d`) ship on metadata only and no numbers are attributed to them. All DOI links resolve to the publisher of record; abstracts remain the copyright of their respective publishers.